

Beyond crisis: The critical role of business continuity management in rebuilding tourism operators' competitiveness

Quratulain Syahirah Awang Ali^a, Mohd Hafiz Hanafiah^{b,c,*}, Spencer Hedley Mogindol^d

^a Faculty of Applied Social Sciences, Universiti Sultan Zainal Abidin, 21300, Gong Badak Campus, Kuala Terengganu, Malaysia

^b Faculty of Hotel and Tourism Management, Universiti Teknologi MARA, 42300, Puncak Alam, Selangor, Malaysia

^c Institute for Biodiversity and Sustainable Development (IBSD), Universiti Teknologi MARA, 40450 Shah Alam, Selangor, Malaysia

^d Faculty of Hotel and Tourism Management, Universiti Teknologi MARA, 88997 Kota Kinabalu, Sabah, Malaysia

ARTICLE INFO

Keywords:

Tourism
Business continuity management
Organizational resilience
Small-scale tour operators

ABSTRACT

This study explores the relationship between Business Continuity Management (BCM) practices and organizational adaptive and planned resilience in small-scale tour operator companies. The study used the multistage sample strategy to collect data from 331 registered tour operator businesses. The empirical findings show that BCM practices, particularly management support, substantially affect both planned and adaptive resilience. However, external requirements and organizational preparedness do not enhance adaptive resilience due to the unpredictable nature of crises. Additionally, planned resilience bridges the gap between BCM practices and adaptive resilience, emphasizing its importance in preparing organizations for unforeseen events. This study highlights the importance of integrating BCM concepts into small-scale organizational culture and strategic decision-making. It offers practical insights for small-scale tourism operators, highlighting the need to emphasize leadership and proactive planning as central to resilience-building, and extends the theoretical understanding of how small-scale tourism businesses can prepare for unforeseen crises in the future.

1. Introduction

Tourism is a key economic sector for many developing countries, including Malaysia, where it contributes significantly to GDP, employment, and regional development. Prior to the COVID-19 pandemic, Malaysia's tourism industry generated RM86.1 billion in revenue in 2019 and welcomed over 26 million international tourists (Tourism Malaysia, 2020). However, the onset of the pandemic brought the global tourism economy to a near standstill. As nations implemented strict border controls and travel bans to contain the virus (Arlou, 2022), international arrivals plummeted by more than 83 % in Malaysia, with tourism revenues plunging to RM12.69 billion in 2020 (Tourism Malaysia, 2020). This economic freefall resulted in widespread business closures, job losses, and disruptions across the tourism supply chain, underscoring the sector's vulnerability to external shocks. While these figures provide stark evidence of tourism's exposure to crisis events, they also reinforce the urgent need for robust Business Continuity Management (BCM) strategies tailored to the sector's specific characteristics.

Despite the heightened attention to organizational resilience in

tourism studies, there is a lack of focused empirical research examining how BCM practices specifically influence the resilience of small-scale tourism businesses. Existing literature often discusses resilience as a broad organizational trait, emphasizing psychological, structural, or community aspects (Duchek, 2020; Prayag et al., 2024). However, few studies delve into how operational frameworks like BCM directly shape planned and adaptive resilience in tourism contexts. Furthermore, studies in more regulated sectors such as finance, healthcare, and logistics highlight the critical role of BCM in maintaining continuity and mitigating risks (Ali et al., 2023; Marcuzzan et al., 2022), but such frameworks are rarely tailored to tourism SMEs, where informal decision-making, resource scarcity, and limited crisis training are the norm.

This gap is particularly pronounced in Southeast Asia, where tourism development varies widely in scale, formality, and institutional support. Although governments have introduced tourism recovery strategies and digital transformation grants, there remains a disconnect between policy-level interventions and ground-level adoption of continuity measures (Rodrigues et al., 2023). Notably, Organisation for Economic Co-operation and Development (OECD) (2022) advocates for MSME

* Corresponding author at: Faculty of Hotel and Tourism Management, Universiti Teknologi MARA, 42300, Puncak Alam, Selangor, Malaysia.

E-mail address: hafizhanafiah@uitm.edu.my (M.H. Hanafiah).

<https://doi.org/10.1016/j.tmp.2025.101418>

Received 28 November 2024; Received in revised form 15 September 2025; Accepted 6 October 2025

Available online 23 October 2025

2211-9736/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

resilience planning through tourism digitalization. However, there is limited insight into how these principles translate into practice within small tourism enterprises, especially during prolonged crises like COVID-19. As a result, understanding the mechanisms through which small-scale tour operators in Malaysia adopt BCM and how such practices influence their resilience strategies is both timely and critical.

This study addresses a dual-pronged problem: the limited empirical understanding of how BCM practices function in small-scale tourism contexts, and the insufficient theorization of how planned resilience mediates the relationship between BCM and adaptive capacity. While prior research acknowledges that BCM enhances performance and post-crisis recovery (Ali et al., 2024), much of this work treats resilience as an outcome rather than a dynamic mediating process (Lexnering & Hollerer, 2022; Mubarik et al., 2022). The literature has largely overlooked how planned resilience—characterized by proactive readiness, scenario planning, and structured risk management—serves as a conduit through which BCM inputs translate into tangible adaptive capabilities, particularly in micro-scale enterprises.

The conceptual model guiding this research is grounded in collective mindfulness theory (Weick & Sutcliffe, 2006), which frames resilience as an organizational process involving sensitivity to emerging threats, commitment to learning, and flexibility in the face of uncertainty. This lens is particularly relevant for small-scale tour operators, who, despite limited resources, possess agility and direct decision-making pathways that allow swift adaptation. By integrating BCM practices—such as management commitment, organizational preparedness, external regulatory compliance, and continuity plan embedding—within a mindfulness framework, the study advances resilience as an active, iterative process rather than a static capacity.

This study thus makes three key contributions. First, it extends empirical investigation of BCM practices to the under-researched segment of Malaysian small-scale tour operators, generating context-specific insights relevant to other developing tourism economies. Second, it clarifies the mechanistic role of planned resilience as a mediating construct, thereby unpacking how BCM influences adaptive resilience. Third, it contributes to applied tourism resilience literature by operationalizing process-driven measures of continuity, resilience, and adaptation—moving the discourse from abstract rhetoric to actionable strategy.

2. Literature review

2.1. Business continuity management (BCM)

Business Continuity Management (BCM) is a practice focused on ensuring that organizations can maintain operations during and after a crisis. Ali et al. (2023) defined BCM as anticipating unfavorable events that may disrupt an organization's operations and implementing strategies to ensure continuity during these disruptions. BCM is critical in risk evaluation, preparation before a crisis, operational management during a crisis, and post-crisis recovery (Crask, 2024). Based on Table 1, the most recent research claimed that organizations must maintain resilience by integrating BCM into their daily operations and decision-making processes, ensuring long-term sustainability and minimizing disruptions caused by unexpected crises. (See Fig. 1.)

2.2. Past studies and hypotheses development

Planned resilience involves proactive measures such as scenario planning, risk assessments, and the establishment of contingency plans, all of which are integral components of Business Continuity Management (BCM) (Widianti et al., 2024). In organizations where BCM is well-embedded, the involvement of senior management and employees plays a crucial role in ensuring that resilience strategies are consistently practiced, evaluated, and refined (Ali et al., 2023). BCM supports planned resilience by ensuring all stakeholders know the potential risks

Table 1
BCM Practices.

BCM Practices	Description	Sources
Management Support	Management support is crucial for effective BCM implementation. It guides decision-making, financial investments, and non-financial resource allocations. Strong commitment from top management increases employee engagement in BCM practices. Leadership's understanding and positive attitudes towards BCM drive organizational growth and crisis management preparedness.	Ali et al. (2024); Boudi and Alshaikhmubarak (2024); Cao et al. (2022); Samsuddin (2023)
External Requirements	External pressures from government policies, competitors, and business partners push organizations to implement BCM. Regulators, clients, and partners demand continuity planning. In tourism, supply chain interdependence drives the need for resilience. External forces also shape sustainability efforts by requiring companies to reassess operational models.	Baloch et al. (2022); Cartwright et al. (2022); Cicerelli and Ravetti (2024); Demir and Demir (2024)
Organizational Preparedness	Strategies and initiatives for risk mitigation ensure crisis readiness. Preparedness includes crisis planning, emergency teams, and BCM protocol enforcement. Organizations with strong preparedness recover more swiftly from disruptions. Employee attitudes and resource allocation influence long-term resilience.	Ali et al. (2023); Ewertowski et al. (2024); Skouloudis et al. (2023)
Embeddedness of Continuity Practices	Effective BCM integration into culture and operations requires awareness campaigns, training, and regular updates. Stakeholder commitment ensures systematic execution and clear recovery plans. Documented BCM processes enhance risk mitigation and resilience.	Ali et al. (2024); Boudi and Alshaikhmubarak (2024); Widianti et al. (2024),
Organizational Resilience	Organizational resilience refers to anticipating, responding to, and recovering from disruptions. It balances flexibility and stability, allowing organizations to navigate uncertainty and adapt without significant structural changes. Ensuring sustainability and business continuity is a core aspect of resilience.	Lexnering and Hollerer (2022); Mandić et al. (2025); Mubarik et al. (2022); Sharma et al. (2021)
Planned Resilience	Proactive measures help mitigate crises before they occur. It includes risk management and BCM strategies to maintain	Bartuseviciene et al. (2024); Sobaih et al. (2021); de Vries and Hamilton (2021)

(continued on next page)

Table 1 (continued)

BCM Practices	Description	Sources
Adaptive Resilience	competitiveness. Pre-disaster planning enhances strategic responses, ensuring smoother recovery and learning from past disruptions.	
	Adaptive resilience is the ability to adjust and respond to crises. Rapid adaptation depends on collaboration and leadership. Small enterprises, especially in tourism, struggle without external support. It ensures business continuity by reconfiguring operations to fit new environments. Social and environmental adaptations enhance long-term sustainability.	Liu-Lastres and Cahyanto (2023); Prayag et al. (2024); Rejan and Ahn (2024); Woodruff et al. (2022)

Source: Researcher compilations.

and are prepared to respond effectively when a crisis arises.

Ali et al. (2024) argue that industries such as healthcare and banking, which are highly regulated, often implement stringent BCM practices that directly influence their ability to build planned resilience. Furthermore, organizational preparedness, management support, and external stakeholder requirements significantly enhance an organization's ability to create a robust BCM framework (Russo et al., 2024). This suggests that when top management actively supports BCM initiatives and aligns them with external expectations, the organization's planned resilience is strengthened, ensuring a faster and more effective recovery from crises (Nguyen et al., 2023). Therefore, it is hypothesized that aspects of BCM, including management support, external requirements, organizational preparedness, and embeddedness of continuity practices, positively influence planned resilience in the tourism sector. The proposed hypotheses are as follows:

H_{1a}: Practicing business continuity through management support influences planned resilience

H_{1b}: Practicing business continuity through external requirements influences planned resilience

H_{1c}: Practicing business continuity through organizational preparedness influences planned resilience

H_{1d}: Practicing business continuity through the embeddedness of continuity practices influences planned resilience

While planned resilience focuses on proactive measures before a crisis, adaptive resilience deals with the organization's ability to respond dynamically during and after the crisis. Adaptive resilience allows organizations to adjust operations, collaborate internally and

externally, and leverage leadership to navigate disruptions effectively (Liu-Lastres & Cahyanto, 2023). BCM plays a critical role in building adaptive resilience by equipping organizations with the necessary tools to respond promptly to unforeseen events. In the tourism sector, external pressures from stakeholders, such as customers, government regulators, and business partners, often drive the need for adaptive resilience (Sawalha, 2020).

Organizations that successfully integrate BCM practices into their culture become more resilient and capable of adjusting their strategies to align with external demands and ensure continuity during crises (Sawalha & Anchor, 2024). For instance, Kato and Charoenrat (2018) and Liu-Lastres and Cahyanto (2023) found that organizations that engaged in comprehensive crisis planning, including allocating resources and preparing crisis teams, were better able to adapt when facing disruptions. Improving crisis management and preparedness strategies through BCM supports adaptive resilience, helping organizations navigate disruptions more flexibly and efficiently (Carty, 2020). This leads to the hypothesis that BCM practices directly influence the adaptive resilience of Malaysian tour operators. The proposed hypotheses are as follows:

H_{2a}: Practicing business continuity through management support influences adaptive resilience

H_{2b}: Practicing business continuity through external requirements influences adaptive resilience

H_{2c}: Practicing business continuity through organizational preparedness influences adaptive resilience

H_{2d}: Practicing business continuity through the embeddedness of continuity practices influences adaptive resilience

Planned resilience is often the foundation for adaptive resilience, as organizations that engage in comprehensive crisis planning and preparation are better equipped to adapt to their operations during a crisis. According to Eichholz et al. (2024) and Thirumaran et al. (2021), effective crisis planning ensures that resources are used efficiently during a disruption and enhances the organization's ability to implement adaptive responses. Proactive organizations that have planned and tested their responses are more likely to navigate a crisis successfully (Prayag et al., 2024). Sobaih et al. (2021) also found a positive relationship between planned and adaptive resilience, where organizations with a clear set of priorities and resilience strategies in place could adapt more swiftly to unforeseen challenges. This study builds on this foundation to examine the relationship between planned and adaptive resilience within Malaysian tour operators, hypothesizing that well-developed planned resilience strategies directly enhance an organization's adaptive resilience capabilities. The proposed hypothesis is:

H₃: Planned Resilience significantly influences adaptive resilience.

BCM has played a pivotal role in shaping both planned resilience and adaptive resilience, with planned resilience acting as a mediator between the two (Lee et al., 2013). When organizations implement BCM effectively, they strengthen their planned resilience, allowing for

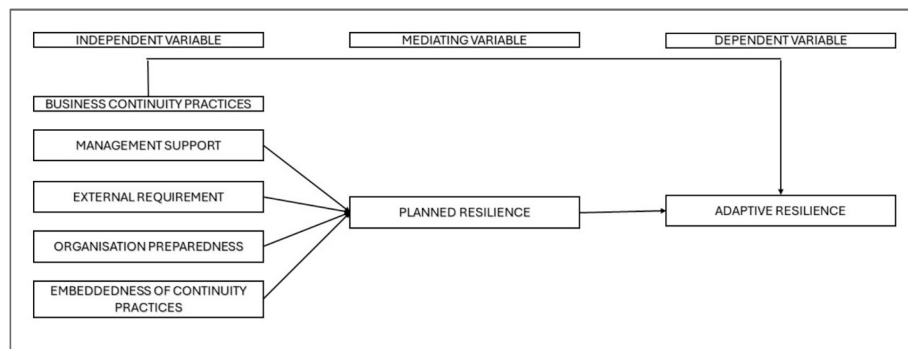


Fig. 1. Proposed Conceptual Framework.

Source: Researcher proposition

structured responses that enable a smooth transition to adaptive resilience during crises (Prayag et al., 2024). Several studies highlight that organizations with strong BCM practices, such as leadership support, proactive risk management, and stakeholder collaboration, are better positioned to develop planned resilience (Ali et al., 2023; Steen et al., 2023). This resilience ensures that organizations are prepared and can adapt quickly to changing circumstances during crises. Järveläinen (2020) emphasizes that BCM enhances adaptive resilience by providing structured responses to disruptions, allowing organizations to recover and adjust operations efficiently.

Planned resilience serves as a bridge by creating the foundation for adaptive resilience. Organizations with robust BCM can leverage planned resilience to ensure flexibility and rapid adaptation during crises (Prayag et al., 2024). Sobaih et al. (2021) further argue that organizations with strong pre-crisis planning can continuously improve their adaptive strategies post-crisis, leading to long-term resilience. In tourism, which faces frequent disruptions, planned resilience facilitates the transition from BCM to adaptive responses, helping organizations maintain continuity and recover quickly (Boin & Van Eeten, 2013). Hence, the proposed hypotheses are:

H_{4a}: Planned Resilience mediates the relationship between management support towards adaptive resilience

H_{4b}: Planned Resilience mediates the relationship between external requirements towards adaptive resilience

H_{4c}: Planned Resilience mediates the relationship between organization preparedness towards adaptive resilience

H_{4d}: Planned Resilience mediates the relationship between the embeddedness of continuity practices towards adaptive resilience

2.3. Underpinning theory

This study adopts Collective Mindfulness Theory (CMT) as the primary theoretical lens to explain how small-scale tourism enterprises develop resilience through Business Continuity Management (BCM) practices. Originating from high-reliability organizations (Weick, 1988; Weick & Sutcliffe, 2006), CMT conceptualizes resilience not as a static trait but as a dynamic, ongoing process marked by heightened awareness, real-time responsiveness, and continuous learning. These characteristics are particularly salient in tourism microenterprises, which often operate with limited formal infrastructure but maintain close managerial oversight and adaptability.

CMT emphasizes five core principles: preoccupation with failure, reluctance to simplify, sensitivity to operations, commitment to resilience, and deference to expertise. This study maps these principles onto key BCM dimensions—management support, external regulatory pressure, organizational preparedness, and embeddedness of continuity practices—arguing that these represent tangible expressions of collective mindfulness in action. For instance, organizational preparedness aligns with preoccupation with failure, while embeddedness of continuity reflects a sustained commitment to resilience and adaptive learning (Ye et al., 2024).

More critically, the study introduces planned resilience as a mediating mechanism that operationalizes mindfulness within the organizational setting. Planned resilience encompasses anticipatory practices such as early warning detection, scenario planning, and structured contingency management, mirroring the proactive, reflective mindset that CMT encourages. The strong relationships observed between BCM constructs and planned resilience support the notion that mindful organizations are those that embed continuity principles into routine operations. This aligns with Reina et al. (2022) and Rusdi and Wibowo (2022), who argue that collective mindfulness strengthens organizational resilience when integrated with formal continuity management systems. Although mindfulness is not measured directly in this study, its dimensions are indirectly expressed through the way BCM practices are implemented, reinforcing the argument that mindfulness can be empirically understood as a process enacted through managerial

behavior and organizational routines.

3. Study methodology

3.1. Study context and design

This study adopted a cross-sectional survey design grounded in a positivist paradigm, appropriate for capturing a snapshot of organizational perceptions and practices during the COVID-19 pandemic. The target population comprised decision-makers from Malaysian small-scale tour operator companies, defined as businesses with fewer than 15 full-time employees consistent with Malaysia's SME classification (SME Corporation Malaysia, 2025). These small-scale operators represent the majority of licensed tourism businesses in Malaysia and play a pivotal role in the sector's resilience and recovery (Wafi et al., 2025).

Sample size was determined using G*Power, requiring a minimum of 133 cases per cluster based on a medium effect size (0.15), statistical power of 0.95, and alpha level of 0.05 (Jobst et al., 2023; Lakens, 2022; Langenberg et al., 2023). A multistage sampling approach was employed. Purposive sampling first identified qualified respondents, those in top management roles directly involved in continuity planning verified via screening questions and follow-up phone calls. Subsequently, cluster sampling was applied using Malaysian states as clusters to ensure regional representation. This combined method addressed potential sampling bias and strengthened generalizability (Dever, 2018; Sharma, 2017).

3.2. Data collection and analysis

Data was collected via the Google Survey platform between October 2022 and February 2023. Survey links were distributed through industry associations, business networks, and direct email invitations. To mitigate self-selection and nonresponse bias, the research team employed follow-up calls, respondent screening, and a bilingual format (Malay and English) to enhance clarity and accessibility. These strategies were particularly important for capturing responses from a diverse pool of small-scale operators, including those with limited digital literacy.

The survey instrument, adapted from Abu Bakar et al. (2015) and Ali et al. (2023), employed a five-point Likert scale (1 = strongly disagree to 5 = strongly agree) and measured constructs related to BCM, planned resilience, and adaptive resilience (see Appendix 1). A pre-test with five tourism experts ensured content validity, while a pilot test involving 30 small-scale operators evaluated reliability and usability. Based on feedback, minor revisions were made. Cronbach's alpha values (0.82–0.94) confirmed internal consistency.

Next, Partial Least Squares—Structural Equation Modeling (PLS-SEM) using SmartPLS version 4.0 was utilized. This study focuses on assessing the reflective measurement model (indicator reliability, internal consistency, discriminant validity) and the structural measurement model (coefficients of determination (R^2), effect size (f^2), predictive relevance (Q^2)). Hair Jr. et al. (2020) and Hanafiah (2020) note that this method is effective in social sciences and tourism management research because of its predictive capabilities and causal explanations.

4. Analysis and results

4.1. Demographic profile

Based on the findings, out of the 350 respondents, only 331 sent their feedback via Google Forms. No "Missing Data" was reported as every question in the survey was compulsory to be answered by each respondent. Generally, Sabah, Wilayah Persekutuan Kuala Lumpur, and Selangor represented 21.1 % ($n = 70$), 17.5 % ($n = 58$), and 17.2 % ($n = 57$) respondents, respectively, which showed the highest percentage out of the 331 respondents. The awareness of BCM among the respondents

was assessed to see the general knowledge of the top management of tour operator companies in Malaysia. It was believed that the BCM perspective is still a new approach that needs to be developed within the tourism industry. Hence, understanding their general awareness towards BCM is significant for the findings of this research. Findings presented that 51.7 % ($n = 171$) of all were totally aware of the BCM concept, while 48.3 % ($n = 160$) were unaware of the BCM concept.

4.2. Measurement model

The measurement model was evaluated for internal consistency, indicator reliability, convergent validity, and discriminant validity, as shown in Table 2 and Fig. 2 (Hanafiah, 2020; Sarstedt et al., 2021). Management Support (MS) and External Requirements (ER) demonstrated exceptional properties, with outer loadings near or above 0.90, CR values of 0.978 and 0.990, and AVE values of 0.896 and 0.949. Organizational Preparedness (OP) had slightly lower outer loadings and CR (0.951) but robust AVE and Cronbach's alpha values (0.811 and 0.940). Embeddedness of Continuity of Practices (ECP) and Planned Resilience (PR) maintained high outer loadings and CR values above 0.95. Adaptive Resilience (AR) showed solid reliability and convergent validity, supporting the suitability of all constructs for further analysis.

Internal consistency reliability was evaluated using Cronbach's alpha and composite reliability (CR), both of which exceeded the 0.70 threshold across all constructs. Convergent validity was confirmed through Average Variance Extracted (AVE) values above 0.50.

Assessing discriminant validity is essential for analyzing interactions between latent variables (Hair Jr. et al., 2013). In variance-based structural equation modeling, this is often tested using the Heterotrait-Monotrait Ratio (HTMT) in PLS-SEM. Henseler et al. (2009) argue that HTMT is superior to traditional methods like the Fornell-Larcker

criterion and cross-loadings, offering more accurate and sensitive results. The HTMT values (see Table 3) are below the 0.90 thresholds, confirming no issues with discriminant validity and effective construct differentiation (Henseler et al., 2015).

This study also assesses the multicollinearity among predictor variables through the Variance Inflation Factors (VIFs). VIF analysis revealed elevated values for External Requirements ($ER = 5.906$) and Management Support ($MS = 4.593$), indicating potential multicollinearity. While values above 5 suggest moderate multicollinearity, they remain below the widely accepted upper threshold of 10 for social sciences (Hair Jr. et al., 2020). Given both constructs' theoretical distinction and empirical significance, they were retained in the model. ER and MS represent conceptually unique dimensions of BCM—external regulatory or market-driven pressures versus internal leadership commitment—which are essential to understanding resilience in tourism businesses. To mitigate potential redundancy, the constructs were rigorously tested for discriminant validity ($HTMT < 0.90$), and the results confirmed that ER and MS measure distinct latent constructs.

4.3. Structural model assessment

After establishing a reliable measurement model, the next step involved analyzing the relationships between independent (exogenous) and dependent (endogenous) latent variables. This analysis tested the study's hypotheses by evaluating the coefficient of determination (R^2), path coefficients (β), effect size (f^2), and predictive relevance (Q^2) (Hair Jr. et al., 2021). The SmartPLS algorithm was used to examine these relationships, with significance determined through regression coefficients (β) and their t-values from the PLS Bootstrapping procedure. Table 4 presents the results of the structural model assessment.

The path analysis reveals that MS, ER, OP, and ECP significantly affect PR. MS has a strong positive impact ($\beta = 0.544^{***}$, $t = 6.127$), ER has a moderate negative effect ($\beta = -0.422^{***}$, $t = 3.637$), and OP ($\beta = 0.229^{**}$, $t = 2.802$) and ECP ($\beta = 0.260^{**}$, $t = 3.302$) also positively contribute. The model accounts for 30 % of PR variance ($R^2 = 0.300$) with a Q^2 of 0.255, indicating good predictive relevance. MS has the most significant impact among the variables, with effect sizes (f^2) ranging from 0.036 to 0.092.

MS and ECP significantly affect AR ($\beta = 0.532^{***}$, $t = 4.928$; $\beta = 0.179^{**}$, $t = 2.933$, respectively), while ER ($\beta = 0.010$, $t = 0.095$) and OP ($\beta = 0.097$, $t = 1.385$) is not a significant predictor for AR. PR also positively impacts AR ($\beta = 0.180^{***}$, $t = 4.448$). This model explains 70.1 % of AR variance ($R^2 = 0.701$) with a Q^2 of 0.543, showing strong predictive relevance. MS has the largest impact on AR ($f^2 = 0.188$), with ECP and PR also contributing ($f^2 = 0.055$ and 0.076), while ER and OP's effects are minimal.

In addition, MS, ER, OP, and ECP significantly influence AR through the PR mediating effect. MS has a strong positive effect ($\beta = 0.098^{***}$, $t = 3.869$), ER shows a slight negative influence ($\beta = -0.076^{**}$, $t = 2.968$), and both OP ($\beta = 0.041^{**}$, $t = 2.446$) and ECP ($\beta = 0.047^{**}$, $t = 2.546$) positively impact AR via PR, though with smaller effect sizes ($f^2 > 0.002$). All pathways are significant, highlighting the role of the independent variables in predicting AR through PR (mediating variable).

5. Study discussion

5.1. BCM influences planned resilience

This study confirms that all proposed BCM components significantly affect planned resilience, aligning with findings by Lee et al. (2013) and Sobaih et al. (2021). While prior research broadly identifies key enablers of resilience—such as simulation exercises, top management commitment, and resource allocation (Settembre-Blundo et al., 2021)—this study offers a more granular view specific to small-scale tourism enterprises. Unlike larger firms that often depend on standardized crisis protocols, small tourism operators demonstrated that flexibility,

Table 2
Measurement Model Assessment.

Construct/ Description	Indicators	Outer Loadings	Composite Reliability	AVE	Cronbach Alpha
Management Support (MS)	MS1	0.912	0.978	0.896	0.977
	MS2	0.928			
	MS3	0.973			
	MS4	0.975			
	MS5	0.974			
	MS6	0.914			
External Requirements (ER)	ER1	0.938	0.990	0.949	0.989
	ER2	0.986			
	ER3	0.959			
	ER4	0.989			
	ER5	0.989			
	ER6	0.982			
Organizational Preparedness (OP)	OP1	0.754	0.951	0.811	0.940
	OP2	0.938			
	OP3	0.957			
	OP4	0.945			
	OP5	0.892			
Embeddedness of Continuity (ECP)	ECP1*		0.968	0.883	0.967
	ECP2	0.921			
	ECP3	0.949			
	ECP4	0.965			
	ECP5	0.943			
	ECP6	0.919			
	ECP7*				
Planned Resilience (PR)	PR1	0.946	0.955	0.872	0.951
	PR2	0.936			
	PR3	0.946			
	PR4*				
	PR5	0.907			
Adaptive Resilience (AR)	AR1	0.845	0.938	0.788	0.933
	AR2	0.909			
	AR3	0.854			
	AR4	0.937			
	AR5	0.891			

Notes: $N = 331$, *deleted.

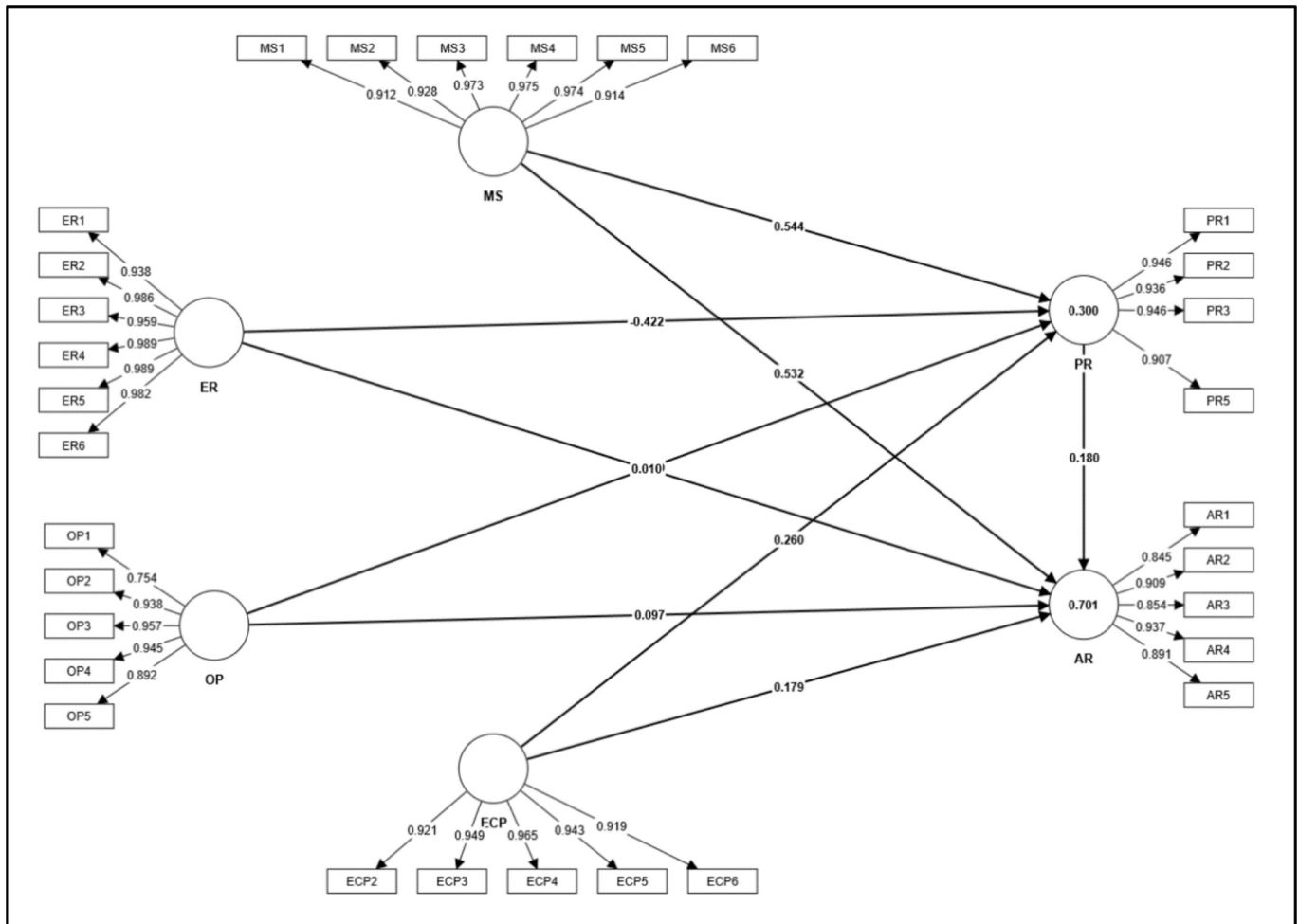


Fig. 2. Measurement Model Result.
Source: SmartPLS output

Table 3
HTMT result.

	AR	ECP	ER	MS	OP
AR					
ECP	0.664				
ER	0.761	0.670			
MS	0.825	0.601	0.889		
OP	0.650	0.605	0.722	0.654	
PR	0.581	0.449	0.399	0.482	0.456

Table 4
Structural Model Assessment Result.

	Path Analysis	β	S.E.	T-Statistics	P-Values	f^2	R^2	Q^2	Results
H _{1a}	MS -> PR	0.544***	0.089	6.127	0.000	0.092	0.300	0.255	Significant
H _{1b}	ER -> PR	-0.422***	0.116	3.637	0.000	0.043			Significant
H _{1c}	OP -> PR	0.229**	0.082	2.802	0.005	0.036			Significant
H _{1d}	ECP -> PR	0.260**	0.079	3.302	0.001	0.052			Significant
H _{2a}	MS -> AR	0.532***	0.108	4.928	0.000	0.188	0.701	0.543	Significant
H _{2b}	ER -> AR	0.010	0.100	0.095	0.924	0.000			Insignificant
H _{2c}	OP -> AR	0.097	0.070	1.385	0.166	0.015			Insignificant
H _{2d}	ECP -> AR	0.179**	0.061	2.933	0.003	0.055			Significant
H ₃	PR -> AR	0.180***	0.040	4.448	0.000	0.076			Significant
H _{4a}	MS -> PR -> AR	0.098***	0.025	3.869	0.000				Significant
H _{4b}	ER -> PR -> AR	-0.076**	0.026	2.968	0.003				Significant
H _{4c}	OP -> PR -> AR	0.041**	0.017	2.446	0.014				Significant
H _{4d}	ECP -> PR -> AR	0.047**	0.018	2.546	0.011				Significant

Notes: **p<.05, ***p<0.001.

improvisation, and informal planning structures were integral to resilience-building. Such complexity challenges the traditional assumption that planned resilience must derive from rigid and pre-established systems.

Moreover, findings show that external influences—such as government recovery strategies—played a more indirect role than expected. Rather than relying on top-down institutional support, many businesses adapted by rapidly pivoting their service models, leveraging digital channels, and engaging in ad-hoc collaborations. This contrasts with Russo et al. (2024), who stress the formal integration of continuity

planning in organizational routines as central to resilience. In small-scale tourism, planned resilience emerged from practical responsiveness to local realities rather than formal policy alignment. Similar patterns have been observed in small retailers during COVID-19, where business survival depended more on digital improvisation and peer learning than on regulatory compliance (Riglietti et al., 2022).

Through the lens of collective mindfulness theory, these findings emphasize that planned resilience in micro-tourism enterprises reflects key mindfulness attributes: sensitivity to operations (local adaptation), reluctance to simplify (non-linear decision-making), and preoccupation with failure (continuous alertness). Hence, planned resilience is not merely a structural input but a mindful process shaped by constant engagement with uncertain environments.

5.2. BCM influences adaptive resilience

The study found that management support and continuity plan embeddedness significantly influence adaptive resilience, confirming previous findings by Chowdhury et al. (2018), Lee et al. (2013), and Madi Odeh et al. (2023). These elements align with CMT's emphasis on deference to expertise and commitment to resilience, as top-down support and integrated planning enable responsive adaptation during crises (Torres & Augusto, 2021; Weick et al., 1999). These results resonate with findings in the healthcare sector, where hospital teams exhibiting collective mindfulness demonstrated greater adaptability in the face of overwhelming demand during pandemics (Weick & Sutcliffe, 2006; Ye et al., 2024).

Conversely, external requirements and organizational preparedness did not significantly affect adaptive resilience in this study, a divergence from expectations based on Duchek (2020). This may reflect the limitations of formalized planning in resource-constrained micro-enterprises, where real-time improvisation and intuitive decision-making often take precedence. Unlike large organizations where preparedness frameworks are regularly rehearsed and resourced, small operators may find such plans outdated or impractical during actual disruptions.

This finding challenges assumptions from Sobaih et al. (2021) and Zhang et al. (2024), who assert that all forms of emergency planning improve adaptive resilience. Instead, the results underscore the contextual limits of top-down continuity frameworks in low-resource environments. This aligns with CMT's critique of over-reliance on standardized procedures, which may reduce situational sensitivity in volatile conditions. For small businesses, adaptive resilience stems more from internal agility and informal collective learning than compliance with rigid protocols.

5.3. Planned resilience influences adaptive resilience

Planned resilience enhanced adaptive resilience significantly, consistent with findings by Sobaih et al. (2021) and Kapucu et al. (2021). However, this study deepens the insight by highlighting how this relationship plays out in businesses that lack formal planning structures. In Malaysian small-scale tourism, planned resilience manifested less as documented emergency plans and more as leadership-driven flexibility and evolving informal strategies.

While Kapucu et al. (2021) link structured planning to improved crisis outcomes, this study reveals that adaptive resilience in small enterprises is contingent on the dynamic application of planned resilience, often shaped by managerial improvisation. This supports the view of Beilstein et al. (2021), who emphasize leadership in crisis recovery. It extends by showing that leadership functions simultaneously as a driver and modifier of resilience strategies throughout the disruption of life-cycles in small firms.

This reinforces CMT's framing of resilience as a real-time, socially distributed process. As Weick et al. (1999) describe, mindfulness emerges not from rigid blueprints but from moment-to-moment sense-making and learning, qualities observed in the adaptive behavior of

small tourism businesses. Therefore, planned resilience in this context should be understood as a fluid, reflexive practice rather than a static crisis protocol.

5.4. Planned resilience mediates BCM practices and adaptive resilience

The mediating role of planned resilience affirms that it serves as a critical bridge between BCM implementation and the development of adaptive capacity, particularly in small-scale enterprises. While Ali et al. (2024) and Widiyanti et al. (2024) underscore the importance of integrating BCM with contingency responsiveness, this study demonstrates that planned resilience must be flexible and customizable to context to function effectively.

Contrary to Nowicki (2024), who views BCM as a standalone planning tool, this study presents evidence that BCM only yields adaptive outcomes when embedded within an evolving resilience mindset. Notably, many tourism operators studied had no formal BCM framework, yet they exhibited high adaptability through experience-based learning and iterative response cycles. These findings align with El-Moumane (2025), who suggests that resilience must extend beyond pre-crisis measures to include in-crisis and post-crisis adaptability.

This argument aligns directly with CMT, which views organizational resilience as emergent, decentralized, and sustained by feedback loops and vigilance. The study thus completes the conceptual loop by showing that BCM practices gain explanatory power only when understood through the process-oriented, socially constructed lens of CMT. It adds empirical weight to the argument that for resilience strategies to work in practice, especially in small tourism firms, they must embody mindful organizing, adaptive responsiveness, and context-aware learning.

6. Study implication

6.1. Theoretical implications

This study advances the theoretical discourse on business continuity management (BCM) and organizational resilience by empirically investigating how simulation exercises, top management engagement, stakeholder trust, and strategic resource deployment contribute to resilience among small-scale tourism operators in Malaysia. While aligning with prior studies emphasizing the role of preparedness in crisis management (Buzzao & Rizzi, 2023; Lee et al., 2013; Sobaih et al., 2021), the findings extend this literature by highlighting a critical shift: regulatory compliance and standardized frameworks alone are not sufficient drivers of adaptive resilience in tourism contexts (Chowdhury et al., 2018; Duchek, 2020).

By challenging the assumed linearity between formal BCM structures and adaptive capacity, this study contributes to a better understanding of the resilience context based on the collective mindfulness theory. Specifically, it demonstrates that internal organizational dynamics—particularly leadership commitment, real-time decision-making, and scenario-based learning—are more decisive factors in rebuilding competitiveness post-crisis. This implies that resilience is not merely a product of external mandates but emerges from embedded, context-sensitive practices that evolve over time. In doing so, the study also extends the application of collective mindfulness theory (Torres & Augusto, 2021; Weick et al., 1999), traditionally examined in large, complex organizations, to small tourism enterprises. The results affirm that a heightened sensitivity to operational dynamics and swift adaptation to unfolding events can significantly enhance recovery and performance even within constrained environments. This cross-contextual validation provides a valuable theoretical bridge, expanding the scope of collective mindfulness to encompass micro-enterprise resilience in volatile tourism markets.

Moreover, introducing planned resilience as a mediating mechanism between BCM practices and adaptive resilience contributes to a novel conceptual lens. It elucidates how structured preparedness—when

infused with iterative learning and contextual intelligence—translates into sustainable adaptive capacities. This mediating role underscores the need to view BCM not as a static protocol but as a dynamic, continuously evolving process that supports long-term strategic agility. Collectively, these insights refine the understanding of resilience-building in tourism by repositioning BCM as a foundation that must be enriched by experiential learning and proactive leadership. The study, therefore, contributes to theory by reframing BCM from a compliance-based practice to a strategic capability, essential for competitiveness in post-crisis tourism recovery. This perspective aligns with and expands existing resilience theories, offering a more integrated framework for future research on crisis navigation and organizational renewal in tourism settings.

6.2. Practical implications

This study provides practical guidance on how Business Continuity Management (BCM) can serve as a strategic tool for enhancing the resilience and competitiveness of small-scale tourism operators in crisis-prone contexts. However, recognizing the resource limitations of many such businesses—especially in rural or underserved regions—it is important to frame the recommendations in terms of both feasibility and scalability. The following actions are organized into short-term, low-cost initiatives and longer-term, resource-intensive strategies to ensure broader applicability and impact.

This study also proposes that short-term small-scale tour operators can adopt low-cost, contextually appropriate interventions that build foundational resilience. For instance, businesses can begin by developing basic risk registers that identify key threats and outline corresponding response plans in a simplified format. Peer-to-peer knowledge exchange—facilitated through community forums, WhatsApp groups, or informal cluster meetings—can be a powerful tool for sharing crisis experiences, lessons learned, and creative problem-solving methods. Informal tabletop exercises using locally relevant scenarios (e.g., flash floods, transportation disruptions, or last-minute booking cancellations) can also simulate decision-making under pressure, helping teams strengthen coordination. Assigning continuity responsibilities to existing staff members and integrating basic BCM principles into daily operations can promote a culture of preparedness without imposing significant financial or technical burdens.

Over the longer term, operators with greater institutional support or stronger digital infrastructure may consider integrating more structured BCM strategies. These may include simulation-based crisis drills, cross-functional staff training, and periodic scenario testing to foster operational agility. Adopting simple digital communication tools—such as automated customer updates via email or messaging platforms—can enhance trust and responsiveness. More advanced efforts, such as real-time scenario analysis, dynamic risk forecasting, and localized resilience audits, may be developed collaboratively with tourism associations or supported through public-private partnerships. Case benchmarking across other tourism sectors or industries like retail and healthcare can also provide transferable models for resilience planning.

The findings call for a shift beyond compliance-driven models for crisis management professionals and destination planners. Adaptive BCM should become central to tourism resilience strategies by incorporating real-time scenario analysis, dynamic risk forecasting, and localized resilience audits. Case benchmarking against successful operators can offer practical learning pathways for others. However, the application of these recommendations must consider regional disparities. In destinations with underdeveloped tourism infrastructure or fragmented institutional support, BCM adoption may require grassroots-level engagement and external facilitation. Still, the core insight remains transferable: management commitment to continuity planning and institutionalizing resilience practices are foundational to long-term competitiveness in volatile tourism environments, particularly across ASEAN and other crisis-prone regions.

7. Conclusion

This study explored a complex relationship between Business Continuity Management (BCM) practices and organizational resilience among small-scale tour operator companies, focusing on understanding how these practices influence the companies' ability to withstand and adapt to disruptions such as the pandemic. The findings underscore the critical role BCM practices play in enhancing small-scale organizational resilience, demonstrating that a robust implementation of BCM strategies encompassing risk analysis, facility preparation, compliance with external regulations, and employee training can significantly improve a company's capacity to manage crises effectively. However, the study also reveals that while external requirements and organizational preparedness alone may not enhance adaptive resilience due to the unpredictable nature of crises, the overall commitment to BCM practices remains vital for fostering flexibility and effective crisis response. These insights emphasize the need for a proactive and integrated approach to BCM within small-scale tourism businesses.

While this study contributes meaningful insights into the relationship between Business Continuity Management (BCM) practices and resilience among small-scale tour operators, several limitations must be acknowledged. First, cross-sectional research design restricts the ability to assess how resilience and BCM practices evolve over time, especially across different phases of a crisis. Second, although mitigation strategies such as follow-up calls and bilingual access were employed, the online survey method may have excluded operators in rural or digitally underserved areas, potentially skewing results towards more resourceful, connected businesses. This raises concerns about the generalizability of findings to informal, unregistered, or marginalized tourism operators. Additionally, the exclusive reliance on quantitative survey data limits the depth of insight into how operators interpret and apply resilience strategies, particularly in the context of crisis response.

To address these limitations, future research should employ longitudinal designs that can capture how planned and adaptive resilience develop and transform over time, particularly during prolonged or repeated disruptions. Incorporating qualitative methods—such as interviews or case studies—would enrich understanding of the BCM implementation's decision-making processes and allow researchers to explore context-specific barriers and enablers of resilience. Expanding the scope beyond small-scale tour operators to include other tourism sub-sectors, such as accommodation, transport, and integrated travel services, would enhance the transferability of findings across the tourism ecosystem. Finally, adopting mixed-mode or offline data collection strategies would help include underrepresented voices, especially from remote or informal operators, ensuring more equitable and comprehensive insights into crisis preparedness and recovery in the tourism industry.

CRedit authorship contribution statement

Quratulain Syahirah Awang Ali: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohd Hafiz Hanafiah:** Writing – original draft, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Spencer Hedley Mogindol:** Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

Declaration

This research work acknowledges the use of generative AI in the preparation of the manuscript. Specifically, ChatGPT was employed for language refinement. All outputs from the AI tools were critically reviewed, edited, and approved by the authors to ensure accuracy, validity, and alignment with the research objectives.

Declaration of competing interest

Following international publication policy and our ethical obligation as a researcher, we report that we have no conflict of interest.

Acknowledgement

The Ministry of Higher Education of MALAYSIA (FRGS/1/2024/WAS01/UITM/01/1) and Universiti Teknologi MARA (600-RMC/FRGS 5/3 (017/2024)) supports this research under the Fundamental Research Grant Scheme.

The author wishes to express his gratitude to the Universiti Teknologi MARA for awarding the fundamental research grant, which facilitated the development of this manuscript.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

References

- Abu Bakar, Z., Yaacob, N. A., & Udin, M. Z. (2015). Business continuity management factors and organisational performance: A study on the moderating role of its capability. *Journal of Management Information*, 2(3), 5–12. <https://doi.org/10.31580/jmi.v7i1.38>
- Ali, Q. S. A., Hanafiah, M. H., & Mogindol, S. H. (2023). Systematic literature review of business continuity management (BCM) practices: Integrating organisational resilience and performance in small and medium enterprises (SMEs) BCM framework. *International Journal of Disaster Risk Reduction*, 104135. <https://doi.org/10.1016/j.ijdrr.2023.104135>
- Ali, Q. S. A., Hanafiah, M. H., & Mogindol, S. H. (2024). Navigating turbulent waters: The impact of business continuity management (BCM) practices on financial and nonfinancial performance of tour operator companies. *Journal of Contingencies & Crisis Management*, 32(1), Article e12550. <https://doi.org/10.1111/1468-5973.12550>
- Arlou, S. (2022). How has the COVID-19 pandemic affected the development of international tourism and what is the role of the vaccination in the tourism industry recovery? *Journal of Positive School Psychology*, 6(7), 2467–2475.
- Baloch, Q. B., Maher, S., Shah, S. N., Sheeraz, M., Iqbal, N., & Raza, H. (2022). Revitalization of tourism and hospitality sector: Pre-empting pandemics through lessons learned. *Environmental Science and Pollution Research*, 29(55), 83099–83111. <https://doi.org/10.1007/s11356-022-21392-7>
- Bartusciene, I., Butkus, M., & Schiuma, G. (2024). Modelling organizational resilience structure: Insights to assess resilience integrating bounce-back and bounce-forward. *European Journal of Innovation Management*, 27(1), 153–169. <https://doi.org/10.1108/EJIM-04-2022-0180>
- Beilstein, C. M., Lehmann, L. E., Braun, M., Urman, R. D., Luedi, M. M., & Stüber, F. (2021). Leadership in a time of crisis: Lessons learned from a pandemic. *Best Practice & Research. Clinical Anaesthesiology*, 35(3), 405–414. <https://doi.org/10.1016/j.bpa.2020.11.011>
- Boin, A., & Van Eeten, M. J. G. (2013). The resilient organization. *Public Management Review*, 15(3), 429–445. <https://doi.org/10.1080/14719037.2013.769856>
- Boudi, A. A., & Alshaikhmubarak, A. (2024). Business continuity and sustainability in government organisations. *Sustainability*, 16(17), 7503. <https://doi.org/10.3390/su16177503>
- Buzzao, G., & Rizzi, F. (2023). The role of dynamic capabilities for resilience in pursuing business continuity: An empirical study. *Total Quality Management and Business Excellence*, 11(1), 18–25. <https://doi.org/10.1080/14783363.2023.2174427>
- Cao, C., Tong, X., Chen, Y., & Zhang, Y. (2022). How top management's environmental awareness affect corporate green competitive advantage: evidence from China. *Kybernetes*, 51(3), 1250–1279. <https://doi.org/10.1108/K-01-2021-0065>
- Cartwright, S., Liu, H., & Davies, I. A. (2022). Influencer marketing within business-to-business organisations. *Industrial Marketing Management*, 106, 338–350. <https://doi.org/10.1016/j.indmarman.2022.09.007>
- Chowdhury, M., Prayag, G., Orchiston, C., & Spector, S. (2018). Post disaster social capital, adaptive resilience and business performance of tourism organisations in Christchurch, New Zealand. *Journal of Travel Research*, 58(7), 1209–1226. <https://doi.org/10.1177/0047287518794319>
- Cicerelli, F., & Ravetti, C. (2024). Sustainability, resilience and innovation in industrial electronics: A case study of internal, supply chain and external complexity. *Journal of Economic Interaction and Coordination*, 19(2), 343–372. <https://doi.org/10.1007/s11403-023-00396-7>
- Crask, J. (2024). *Business continuity management: A practical guide to organization resilience and ISO 22301*. Kogan Page Publishers.
- Demir, M., & Demir, Ş.Ş. (2024). Is the global technological outage a caution for the service industries? Evidence from the tourism industry. *Current Issues in Tourism*, 1–21. <https://doi.org/10.1080/13683500.2024.2408581>
- Dever, J. (2018). Combining probability and nonprobability samples to form efficient hybrid estimates: An evaluation of the common support assumption. In *Proceedings of the 2018 Federal Committee on statistical methodology (FCSM) research conference* (pp. 1–15).
- Duchek, S. (2020). Organisational resilience: A capability-based conceptualization. *Business Research*, 13(1), 215–246. <https://doi.org/10.1007/s40685-019-0085-7>
- Eichholz, J., Hoffmann, N., & Schwering, A. (2024). The role of risk management orientation and the planning function of budgeting in enhancing organizational resilience and its effect on competitive advantages during times of crises. *Journal of Management Control*, 35(1), 17–58. <https://doi.org/10.1007/s00187-024-00371-8>
- El-Moumane, R. (2025). 433. *The role of strategic planning in resilience and crisis management: A case study of Enhancing Resilience in Business Continuity Management*.
- Ewertowski, T., Güldoğan, B.Ç., Kuter, S., Akçöz, S., Weber, G. W., Sadiowska-Wrzesińska, J., & Racek, E. (2024). The use of machine learning techniques for assessing the potential of organizational resilience. *Central European Journal of Operations Research*, 32(3), 685–710. <https://doi.org/10.1007/s10100-023-00875-z>
- Hair, J. F., Jr., Howard, M. C., & Nitzl, C. (2020). Assessing measurement model quality in PLS-SEM using confirmatory composite analysis. *Journal of Business Research*, 109, 101–110. <https://doi.org/10.1016/j.jbusres.2019.11.069>
- Hair, J. F., Jr., Hult, G. T. M., Ringle, C. M., Sarstedt, M., Danks, N. P., & Ray, S. (2021). *Partial least squares structural equation modelling (PLS-SEM) using R: A workbook*. Springer Nature.
- Hair, J. F., Jr., Ringle, C. M., & Sarstedt, M. (2013). Partial least squares structural equation modeling: Rigorous applications, better results and higher acceptance. *Long Range Planning*, 46(1–2), 1–12. <https://doi.org/10.1016/j.lrp.2013.01.001>
- Hanafiah, M. H. (2020). Formative vs. reflective measurement model: Guidelines for structural equation modeling research. *International Journal of Analysis and Applications*, 18(5), 876–889. <https://doi.org/10.28924/2291-8639-18-2020-876>
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy Marketing*, 43, 115–135. <https://doi.org/10.1007/s11747-014-0403-8>
- Henseler, J., Ringle, C. M., & Sinkovics, R. R. (2009). The use of partial least squares path modeling in international marketing. *New Challenges to International Marketing*, 20, 277–320. [https://doi.org/10.1108/S1474-7979\(2009\)0000020014](https://doi.org/10.1108/S1474-7979(2009)0000020014)
- Järveläinen, J. (2020). Understanding the stakeholder roles in business continuity management practices – A study in public sector. In *Proceedings of the 53rd Hawaii international conference on system sciences*. <https://doi.org/10.24251/hicss.2020.241>
- Jobst, L. J., Bader, M., & Moshagen, M. (2023). A tutorial on assessing statistical power and determining sample size for structural equation models. *Psychological Methods*, 28(1), 207. <https://doi.org/10.1037/met0000423>
- Kapucu, N., Hu, Q., Harmon, M., & Toro, P. (2021). Coordinating non-established disaster relief groups: a case study of Hurricane Irma in Florida, United States. *Disasters*, 45(3), 717–737. <https://doi.org/10.1111/disa.12439>
- Kato, M., & Charoenrat, T. (2018). Business continuity management of small and medium size enterprises: Evidence from Thailand. *International Journal of Disaster Risk Reduction*, 27, 577–587. <https://doi.org/10.1016/j.ijdrr.2017.10.002>
- Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, 8(1), Article 33267. <https://doi.org/10.1525/collabra.33267>
- Langenberg, B., Janczyk, M., Koob, V., Kliegl, R., & Mayer, A. (2023). A tutorial on using the paired t test for power calculations in repeated measures ANOVA with interactions. *Behavior Research Methods*, 55(5), 2467–2484. <https://doi.org/10.3758/s13428-022-01902-8>
- Lee, A. V., Vargo, J., & Seville, E. (2013). Developing a tool measure and compare organisations resilience. *Natural Hazards Review*, 14(1), 29–41. [https://doi.org/10.1061/\(ASCE\)NH.1527-6996.0000075](https://doi.org/10.1061/(ASCE)NH.1527-6996.0000075)
- Lexnering, S., & Hollerer, M. (2022). Remaining the same or becoming another? Adaptive resilience versus transformative urban change. *Urban Studies*, 59(6), 1300–1310. <https://doi.org/10.1177/0042098021998923>
- Liu-Lastres, B., & Cahyanto, I. P. (2023). Are we always ready? Examining event professionals' approaches to risk and crisis management and resilience. *Tourism Management Perspectives*, 46, Article 101073. <https://doi.org/10.1016/j.tmp.2023.101073>
- Madi Odeh, R. B. S., Obaidat, B. Y., Jaradat, M. O., Masa'deh, R., & Alshurideh, M. T. (2023). The transformational leadership role in achieving organisational resilience through adaptive cultures: The case of Dubai service sector. *International Journal of Productivity and Performance Management*, 72(2), 440–468. <https://doi.org/10.1108/IJPPM-02-2021-0093>
- Mandić, A., Petrić, L., & Pivčević, S. (2025). Harmonizing sustainability and resilience in post-crisis cultural tourism: Stakeholder insights from the Split metropolitan area living lab. *Tourism Management Perspectives*, 55, Article 101331. <https://doi.org/10.1016/j.tmp.2024.101331>
- Marcuzzan, E., Campagnolo, D., & Gianecchini, M. (2022). Reaction or anticipation? Resilience in small- and medium- sized enterprises. *Journal of Small Business and Enterprise Development*, 29(5), 764–788. <https://doi.org/10.1108/JSBED-07-2021-0271>
- Mubarik, M. S., Bontis, N., Mubarik, M., & Mahmood, T. (2022). Intellectual capital and supply chain resilience. *Journal of Intellectual Capital*, 23(3), 713–738. <https://doi.org/10.1108/JIC-06-2020-0206>
- Nguyen, T. H. D., Chileshe, N., Rameezdeen, R., & Wood, A. (2023). Strategic responses to external stakeholder influences. *International Journal of Project Management*, 41(1), Article 102434. <https://doi.org/10.1016/j.ijproman.2022.102434>
- Nowicki, P. (2024). Selected concepts of risk management. In J. Sitko-Lutek, A. Jabłoński, & M. Jabłoński (Eds.), *Quality of the future – The future of quality (QFFQ)* (pp. 148–161). Sciend. <https://doi.org/10.2478/9788367405843-014>

- Organisation for Economic Co-operation and Development (OECD). (2022). *OECD tourism trends and policies 2022*. OECD Publishing. <https://doi.org/10.1787/a8dd3019-en>
- Prayag, G., Jiang, Y., Chowdhury, M., Hossain, M. I., & Akter, N. (2024). Building dynamic capabilities and organizational resilience in tourism firms during COVID-19: A staged approach. *Journal of Travel Research*, 63(3), 713–740. <https://doi.org/10.1177/0047287523116497>
- Reina, C. S., Kreiner, G. E., Rheinhardt, A., & Mihelcic, C. A. (2022). Your presence is requested: Mindfulness infusion in workplace interactions and relationships. *Organisation Science*, 34(1), 12–19. <https://doi.org/10.1287/orsc.2022.1596>
- Rejan, K., & Ahn, Y. J. (2024). Challenges, experiences, and coping behaviors among SMEs in the adventure tourism industry after the COVID-19 pandemic: A case from Pokhara, Nepal. *Journal of Outdoor Recreation and Tourism*, 47, Article 100797. <https://doi.org/10.1016/j.jort.2024.100797>
- Riglietti, G., Piraina, M., & Trucco, P. (2022). The contribution of business continuity management (BCM) to supply chain resilience: a qualitative study on the response to COVID-19 outbreak. *Continuity & Resilience Review*, 4(2), 145–160.
- Rodrigues, V., Eusébio, C., & Breda, Z. (2023). Enhancing sustainable development through tourism digitalisation: a systematic literature review. *Information Technology & Tourism*, 25(1), 13–45.
- Rusdi, Z. M., & Wibowo, A. (2022). The underlying process of high-performance work systems on innovation through collective mindfulness. *Atlantis Press*, 374–379. <https://doi.org/10.2991/aebmr.k.220702.057>
- Russo, N., Reis, L., Silveira, C., & Mamede, H. S. (2024). Towards a comprehensive framework for the multidisciplinary evaluation of organizational maturity on business continuity program management: a systematic literature review. *Information Security Journal: A Global Perspective*, 33(1), 54–72.
- Samsuddin, K. N. (2023). Innovative management reporting for real time dynamic decision making. In *Gas and oil technology showcase and conference, Dubai, UAE*. <https://doi.org/10.2118/214210-MS>
- Sarstedt, M., Ringle, C. M., & Hair, J. F. (2021). Partial least squares structural equation model. In *Handbook of market research*. Springer.
- Sawalha, I. H. (2020). Business continuity management: Use and approach's effectiveness. *Continuity and Resilience Review*, 2(2), 81–96. <https://doi.org/10.1108/CRR-05-2020-0016>
- Sawalha, I. H., & Anchor, J. R. (2024). Interpretations of business continuity management in the light of COVID-19. *Management & Sustainability: An Arab Review*, 3(3), 233–248. <https://doi.org/10.1108/MSAR-01-2023-0004>
- Settembre-Blundo, D., González-Sánchez, R., Medina-Salgado, S., & García-Muiña, F. E. (2021). Flexibility and resilience in corporate decision making: A new sustainability-based risk management system in uncertain times. *Global Journal of Flexible Systems Management*, 22(Suppl. 2), 107–132. <https://doi.org/10.1007/s40171-021-00277-7>
- Sharma, G. (2017). Pros and cons of different sampling techniques. *International Journal of Applied Research*, 3(7), 749–752.
- Sharma, G. D., Thomas, A., & Paul, J. (2021). Reviving tourism industry post-covid-10: A resilience-based framework. *Tourism Management Perspectives*, 37. <https://doi.org/10.1016/j.tmp.2020.100786>
- Skouloudis, A., Leal Filho, W., Deligiannakis, G., Vouros, P., Nikolaou, I., & Evangelinos, K. (2023). Coping with floods: Impacts, preparedness and resilience capacity of Greek micro-small-medium-sized enterprises in flood affected areas. *International Journal of Climate Change Strategies and Management*, 15(1), 81–103. <https://doi.org/10.1108/IJCCSM-09-2022-0122>
- SME Corporation Malaysia. (2025). SME definition. Retrieved April 22, 2025, from <http://smecorp.gov.my/index.php/en/policies/2020-02-11-08-01-24/sme-definition>.
- Sobaih, A. E. E., Elshaer, I., Hasanein, A. M., & Abdelaziz, A. S. (2021). Responses to COVID-19: the role of performance in the relationship between small hospitality enterprises' resilience and sustainable tourism development. *International Journal of Hospitality Management*, 94, 1–23. <https://doi.org/10.1016/j.ijhm.2020.102824>
- Steen, R., Haug, O. J., & Patriarca, R. (2023). Business continuity and resilience management: A conceptual framework. *Journal of Contingencies & Crisis Management*, 32(1). <https://doi.org/10.1111/1468-5973.12501>. Article e12501.
- Thirumaran, K., Mohammadi, Z., Pourabedin, Z., Azzali, S., & Sim, K. (2021). COVID-19 in Singapore and New Zealand: Newspaper portrayal, crisis management. *Tourism Management Perspectives*, 38, Article 100812. <https://doi.org/10.1016/j.tmp.2021.100812>
- Torres, P., & Augusto, M. (2021). Attention to social issues and CEO duality as enablers of resilience to exogenous shocks in tourism industry. *Tourism Management*, 87, 1–12. <https://doi.org/10.1016/j.tourman.2021.104400>
- Tourism Malaysia. (2020). Malaysia tourism statistics in brief. Retrieved Sept 18, 2021 from <https://www.tourism.gov.my/statistics>.
- de Vries, H. P., & Hamilton, R. T. (2021). Smaller businesses and the Christchurch earthquakes: A longitudinal study of individual and organizational resilience. *International Journal of Disaster Risk Reduction*, 56, Article 102125. <https://doi.org/10.1016/j.ijdrr.2021.102125>
- Wafi, A. A., Mohamed, S., Hanapi, Z., Ibrahim, J. A., & Samsudin, N. (2025). Creating a sustainable tourism business model for small-scale tourism products in Malaysia: Embracing tourism opportunity for post-pandemic resilience. *Pertanika Journal of Social Sciences & Humanities*, 33(1).
- Weick, K. E. (1988). Introductory essay improvisation as a mindset for organisational analysis. *Organisation Science*, 9(5), 543–555. <https://doi.org/10.1287/orsc.9.5.543>
- Weick, K. E., & Sutcliffe, K. M. (2006). Mindfulness and the quality of organisational attention. *Organisation Science*, 17(4), 514–524. <https://doi.org/10.1287/orsc.1060.0196>
- Weick, K. E., Sutcliffe, K. M., & Obstfeld, D. (1999). Organizing for high reliability: Processes of collective mindfulness. *Research in Organisational Behaviour*, 21, 81.
- Widianti, T., Dinaseviani, A., Ayundyahrini, M., Mahmudi, M., Susanto, A., Nuryanto, I., ... Suherman, U. (2024). Business continuity management: Trends, structures and future issues. *Business Process Management Journal*, 30(7), 2352–2379. <https://doi.org/10.1108/BPMJ-01-2024-0046>
- Woodruff, S. C., Meerow, S., & Wilkins, C. (2022). Adaptation to resilience planning: Alternative pathways to prepare for climate change. *Journal of Planning Education and Research*, 42(1), 64–75. <https://doi.org/10.1177/0739456X18801057>
- Ye, D., Liu, M. J., Luo, J., & Yannopoulou, N. (2024). How to achieve swift resilience: The role of digital innovation enabled mindfulness. *Information Systems Frontiers*, 26(2), 551–573. <https://doi.org/10.1007/s10796-021-10225-6>
- Zhang, F., Lv, Y., & Sarker, M. N. I. (2024). Resilience and recovery: A systematic review of tourism governance strategies in disaster-affected regions. *International Journal of Disaster Risk Reduction*, 104350.



Quratulain Syahirah Awang Ali completed her PhD in Hotel and Tourism Management from UiTM in 2023. She is currently attached to the Faculty of Applied Social Sciences, Universiti Sultan Zainal Abidin, Terengganu, MALAYSIA. She is an active author focusing on tourism business resilience and business continuity.



Mohd Hafiz Hanafiah is a Professor at the Faculty of Hotel & Tourism Management, Universiti Teknologi MARA, Malaysia. He completed his PhD in Hotel and Tourism Management from UiTM in 2016 and Master in Economics from Universiti Malaya in 2008. He is an active author, authoring multiple research papers in tourism and hospitality management, focusing on tourism destination competitiveness, tourist behavior and technology adoption.



Spencer Hedley Mogindol is a Senior Lecturer at the Faculty of Hotel and Tourism Management, Universiti Teknologi MARA Sabah, Malaysia. He obtained his PhD in Tourism Management from Universiti Malaysia Sabah, Malaysia. He is an active researcher in the community based tourism area.